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SEA: Hierarchically searching efficient adapters for pre-trained models

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ABSTRACT

Adapting large-scale pre-trained models for downstream tasks has proven to be effective in improving performance. However, relying on the hand-crafted adapters may limit generalizability and incur extra inference overhead. Existing neural architecture search (NAS) techniques for designing adapters generally focus on hyperparameters, neglecting the design of the search space and the biased problem in the supernet training. To address these limitations, this paper proposes a general framework to **S**earch **E**fficient **A**dapters (**SEA**), providing a hierarchical adapter search space from a systematic perspective and introducing a sample-efficient supernet training mechanism. Specifically, SEA searches for efficient adapters at various levels of granularity, enabling refined adapter depths, widths, and channel combinations. The hierarchical search space encompasses a diverse range of adapter structures, enabling generalizability across multiple tasks. Furthermore, the searched adapters can be seamlessly integrated into the pre-trained models, eliminating any additional inference overhead. To avoid training bias of different channels during the width search, we implement a sample-efficient training mechanism inspired by the Upper Bound Confidence (UCB) theory. This mechanism promotes equal opportunities for training channels at early stages and prioritizes training of channels with larger gradient norms later, enhancing the training's effectiveness. We provide adaption results on a wide range of tasks, showing the superiority of our method. SEA achieves 0.53% higher average accuracy than GLoRA across 19 few-shot tasks on VTAB-1k. When adapted on COCO, SEA boosts mean average precision by 7.3% (detection) and 5.6% (segmentation) over SSF. Moreover, we delve into the relationship between adapter structures and downstream tasks, unveiling interesting phenomena that may offer insights for future adapter design. Our code is available at https://github.com/ShunLu91/SEA.

1. Introduction

Recent years have witnessed exponential growth in developing large-scale models (Bao et al., 2022; He et al., 2022b; Li et al., 2019; Liu et al., 2019a; Yuan et al., 2021), that have achieved remarkable performance across various tasks. However, due to limited training resources and data availability on downstream tasks, directly fine-tuning large models becomes a challenging endeavor. Consequently, many approaches (Chen et al., 2022; Houlsby et al., 2018; Hu et al., 2022; Jia et al., 2022; Jiang et al., 2023; Lian et al., 2022) resort to a parameter-efficient fine-tuning (PEFT) paradigm. These methods keep the pre-trained weights of the large-scale model frozen while only fine-tuning an efficient adapter, making the model quickly adapt to downstream tasks. Thanks to its exceptional efficiency, such a technique has found widespread applications.

Despite the great progress, existing hand-crafted adapters (Chen et al., 2022; Houlsby et al., 2018; Hu et al., 2022; Jia et al., 2022) not only require extensive expertise but also lack generalizability. Specifically, designing adapter structures and selecting their width (hidden dimension or rank) for different downstream tasks still heavily depends on human experience. Typically, extensive ablation studies are required to identify a suitable configuration, which is both time-consuming and

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resource-intensive. Moreover, the structural requirements of adapters inherently differ across task types due to their distinct characteristics: image classification often relies only on high-level semantic cues and can be effectively adapted with shallow, low-rank adapters, whereas dense prediction tasks like detection and segmentation demand fine-grained spatial reasoning and thus benefit from deeper adapters with higher ranks. Consequently, transferring an adapter across diverse tasks necessitates re-designing its structure from scratch, incurring significant engineering overhead. Moreover, previous adapters such as VPT (Jia et al., 2022) and AdaptFormer (Chen et al., 2022) introduce extra inference overhead, further exacerbating the efficiency of large-scale models.

Due to the incapability of fixed adapters in adapting to diverse downstream tasks, **NAS emerges as a natural choice for the adapter design**. NOAH (Liu, 2022) and GLoRA (Chavan et al., 2023) have pioneered this way. However, the searched adapter in NOAH still incurs extra inference costs. Furthermore, neither method systematically designs the adapter search space nor addresses the difficulties of training different channels during the search. These limitations prompt us to elaborate a hierarchical search space to search for efficient adapters from a systematic perspective in this work.

Our primary principle is to ensure that the searched adapters can be seamlessly integrated into large pre-trained models without introducing any additional inference overhead. This is crucial to maintain both the efficiency and the practicality of the adapter-based approach. Our second principle is to ensure strong cross-task generalization: we achieve this through a hierarchical search space that decouples adapter configuration into depth, width, and channel dimensions, enabling automatic adaptation to diverse task characteristics. This decoupling enables task-aware adaptation-automatically allocating adapter resources according to the intrinsic demands of each downstream task, thereby eliminating manual tuning and improving generalization. By exploring different adapter depths, we allow for the adaptation of different levels of abstractions within the model. For instance, shallow depths suffice for tasks relying on simple semantics (e.g., image classification), while deeper adaptations are beneficial for dense prediction requiring spatial fidelity. The width determines the capacity of the adapters to capture task-specific information since the appropriate width corresponds to better task-specific adaptations. Rather than using a fixed width, our search dynamically assigns per-layer capacity based on task complexity. Besides, the channel combinations enable us to investigate the effectiveness of different channels under the same width to further enhance the adaption performance at a fine-grained level. This fine-grained selection identifies the most informative feature subspaces for adaptation, reducing redundancy and boosting efficiency without increasing rank.

To alleviate biased training for the supernet, we introduce a sample-efficient method adopted from Upper Confidence Bound (UCB) theory (Auer et al., 2002) to balance the training of different channels, leading to fairer comparison between channels and faster training convergence. This mechanism better balances the exploration and exploitation of channels by jointly utilizing the channel sampling frequencies and gradient norms. As the under-explored yet promising channels obtain sufficient training, this approach enhances the fairness of training across channels and improves convergence. Therefore, it manages to search for superior adapters serving for downstream task adaptation.

Our contributions are summarized as follows:

- We introduce a hierarchical search space to enable automated search of adapter depth, width, and channel combinations, facilitating versatile generalization across diverse downstream tasks. The searched adapter can be seamlessly integrated into the original model without introducing additional inference overhead.
- To mitigate biased training for the supernet, we propose a sample-efficient training approach inspired by UCB theory. This method promotes fairness and convergence in training across different channels, enabling rapid adaptation and convergence for the supernet. Moreover, the discovered adapters do not require re-training or fine-tuning, saving computational resources.
- Our experiments span multiple downstream tasks, including transfer learning, few-shot learning, and domain generalization. SEA consistently outperforms state-of-the-art methods with notable improvements. Moreover, we conduct an in-depth analysis of the relationship between the adapters and downstream tasks, offering insights for future adapter design.

2. Related work

2.1. Parameter efficient adapters

Parameter-efficient tuning has become a standard paradigm for adapting large pre-trained vision models without full fine-tuning. Early approaches such as LoRA (Hu et al., 2022) inject trainable low-rank matrices into attention layers, achieving strong performance with minimal overhead. Visual Prompt Tuning (VPT) (Jia et al., 2022) prepends learnable tokens to input sequences, but its effectiveness diminishes in deeper layers or dense prediction tasks. Subsequent works like AdaptFormer (Chen et al., 2022) insert lightweight MLPs into feed-forward blocks, while Res-Tuning (Jiang et al., 2023) introduces residual adapters that preserve original features via skip connections. Despite their success, these methods rely on fixed architectures, requiring manual selection of rank, depth, or placement, which limits adaptability across heterogeneous tasks.

2.2. NAS for adapters

To automate adapter design, recent works apply neural architecture search (NAS). In the field of Natural Language Processing (NLP), AdaLoRA (Zhang et al., 2023) and DyLoRA (Valipour et al., 2023) search for optimal ranks for adapters. In computer vision, NOAH (Liu, 2022) constructs a unified space over hand-crafted adapters to search for per-layer width and adapter type, while GLoRA (Chavan et al., 2023) unifies LoRA, SSF, and RepAdapter (Luo et al., 2023) into a single framework for structure and width selection.

Complementing these efforts, Perrin et al. (2024) emphasize the importance of configurable search spaces in general NAS-a principle that motivates our decoupled design over depth, width, and channel dimensions. Meanwhile, Liang et al. (2026) enhance LoRA's expressiveness via learnable subspace transformations, highlighting that parameterization geometry matters; in contrast, SEA focuses on *automatically selecting* the best adapter architecture rather than refining a fixed one.

However, these methods either fix the adapter depth (i.e., which layers to adapt) or assume a default channel selection (e.g., top-r channels in LoRA). They lack a holistic search space that jointly optimizes *where* (depth), *how much* (width), and *which features* (channel combinations) to adapt. Our work addresses this gap by proposing a hierarchical search space that decouples these three dimensions, enabling task-adaptive adapter discovery. Inspired by ViTAS (Su et al., 2022)'s fair channel training, we further incorporate channel-level exploration to identify superior feature subspaces beyond fixed selections.

3. Method

In this section, we start by revisiting Transformer-based models and the low-rank adaption method. Following that, we introduce our hierarchical adapter search space. We then present our sample-efficient mechanism inspired by UCB theory for the supernet training. Finally, we discuss the application of evolutionary search for efficient adapters.

3.1. Background

Transformer-based Models. Transformer-based models have gained widespread application in various fields, including computer

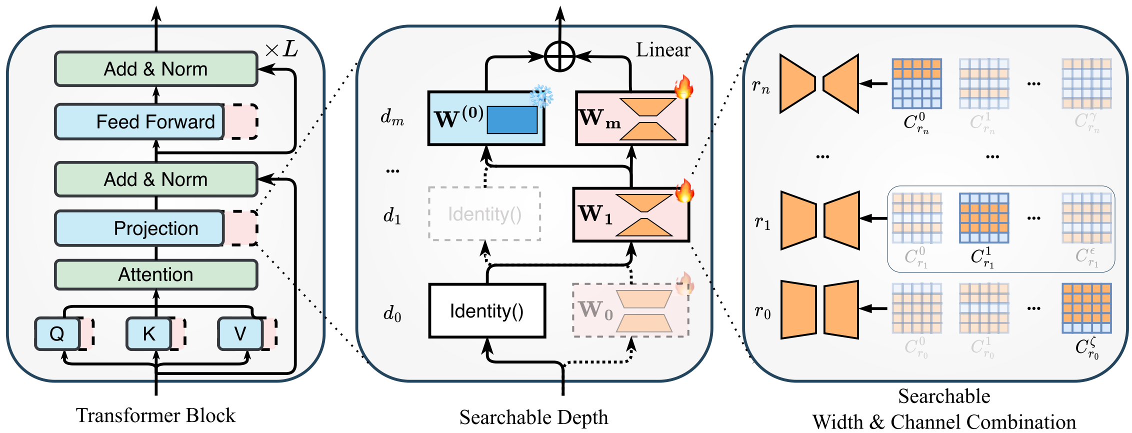


Fig. 1. Our proposed hierarchical transformer-based adapter search space. Left: the fundamental structure of a typical visual transformer. Pink areas enclosed by a dashed line represent the introduction of a trainable adapter, whose hierarchical structures are shown in the following diagrams. Middle: the depth search space of the adapter designed for a specific linear layer. Right: the width and channel combination search space of the adapter. For a candidate width r_i, it can be formed by different channel combinations. The bias of the linear layer and the residual connection are omitted in the diagrams.

vision (Bao et al., 2022; He et al., 2022b), natural language processing (Kenton & Toutanova, 2019; Liu et al., 2019a), and others (Li et al., 2019; Yuan et al., 2021). The backbone of these models usually serially stacks L Transformer layers. Taking the Vision Transformer (Dosovitskiy et al., 2021) as an example, as depicted on the left side of Fig. 1, it mainly comprises a multi-head self-attention (MHSA) module and a feed-forward layer (FFN). Given the input feature $\boldsymbol{x} \in \mathbb{R}^{n\times d}$, it is first mapped to queries $\mathbf{Q} = \boldsymbol{x}\mathbf{W}_q$, keys $\mathbf{K} = \boldsymbol{x}\mathbf{W}_k$, and values $\mathbf{V} = \boldsymbol{x}\mathbf{W}_v$ through the linear layer $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{d\times d}$. Then the MHSA is computed by

$$\boldsymbol{x}_{\text{attn}} = \text{Softmax}(\frac{\mathbf{QK}^\top}{\sqrt{d}})\mathbf{V} \tag{1}$$

After a residual connection, the outputs of the attention module $\boldsymbol{x}_{\text{attn}} \in \mathbb{R}^{n\times d}$ is processed by the FFN:

$$\boldsymbol{x}_{\text{ffn}} = \phi(\boldsymbol{x}_{\text{attn}}\mathbf{W}_{f_1} + \mathbf{b}_{f_1})\mathbf{W}_{f_2} + \mathbf{b}_{f_2} \tag{2}$$

where $\mathbf{W}_{f_1} \in \mathbb{R}^{d\times d_f}$ and $\mathbf{W}_{f_2} \in \mathbb{R}^{d_f\times d}$ are learnable weights, $\mathbf{b}_{f_1} \in \mathbb{R}^{d_f}$ and $\mathbf{b}_{f_2} \in \mathbb{R}^{d}$ are the biases, ϕ denotes the activation function.

Low-Rank Adaption Method. LoRA Hu et al. (2022) proposes to model the incremental update $\Delta\mathbf{W} \in \mathbb{R}^{d\times d}$ of the pre-trained weights $\mathbf{W}^{(0)} \in \mathbb{R}^{d\times d}$ during adaption by a product of two low-rank matrices $\mathbf{A} \in \mathbb{R}^{R\times d}$ and $\mathbf{B} \in \mathbb{R}^{d\times R}$, the resulting forward features h can be expressed as:

$$\boldsymbol{h} = \boldsymbol{x}\mathbf{W}^{(0)} + \boldsymbol{x}\Delta\mathbf{W} = \boldsymbol{x}\mathbf{W}^{(0)} + \boldsymbol{x}\mathbf{BA} \tag{3}$$

Initially, LoRA utilizes the low-rank matrices to model the updates of pre-trained $\mathbf{W}_q$ and $\mathbf{W}_v$. Later, many works have extended this paradigm to all pre-trained weights in the Transformer block. Due to its high efficiency and seamless integration merit, LoRA has been widely applied. In the following, our hierarchical search space design is also based on the low-rank matrices.

3.2. Hierarchical search space for adapters

Although previous NAS works (Chavan et al., 2023; Liu, 2022) have adopted the low-rank matrices, they often focus on searching for the optimal ranks of these matrices, neglecting the systematic design of the search space. In contrast, we introduce a hierarchical search space that explores low-rank matrices across three dimensions: depth, width, and channel combinations, enhancing adaptability to downstream tasks.

Depth. As shown in the middle of Fig. 1, we devise both parallel and serial low-rank matrices $[\mathbf{W}_0, \mathbf{W}_1, \dots, \mathbf{W}_m]$ for each pre-trained weight matrix $\mathbf{W}^{(0)}$, The parallel low-rank matrix serves as a mask to emphasize the important elements of $\mathbf{W}^{(0)}$, while the serial low-rank matrices facilitate improved learning of downstream task features. The resulting forward process of the middle of the Fig. 1 can be expressed as:

$$\begin{cases} \boldsymbol{y}_0 = \text{Identity}(\boldsymbol{x}), \boldsymbol{y}_1 = \boldsymbol{y}_0\mathbf{W}_1 + \mathbf{b}_1 + \boldsymbol{x}, \dots \\ \boldsymbol{y}_m = (\boldsymbol{y}_{m-1}\mathbf{W}^{(0)} + \mathbf{b}^{(0)}) + (\boldsymbol{y}_{m-1}\mathbf{W}^{(0)}\mathbf{W}_m + \mathbf{b}_m\mathbf{b}^{(0)}) \end{cases} \tag{4}$$

In line with the analysis by GLoRA (Chavan et al., 2023), this combined parallel-serial design can capture both the changes in weight-space $\Delta\mathbf{W}$ and the changes in feature-space ΔY, enabling more effective adaptation to downstream tasks.

Width. In the context of the width dimension, our focus lies in searching for the rank of the trainable low-rank matrices. Given n searchable ranks $R = [r_0, r_1, \dots, r_n]$ arranged in ascending order, we choose a rank r_i as the width of the i_{th} adapter weights $\mathbf{W}_i$. This emphasis stems from the recognition that weight-space changes vary across pre-trained weights at different layers or positions. Therefore, it is critical to discover a suitable rank for each pre-trained weight to achieve optimal adaptation to downstream tasks.

Channel Combination. In previous approaches, a selected rank r_i typically leads to the selection of the top-r_i channels to construct the i_{th} adapter weights $\mathbf{W}_i$. This practice restricts the diversity of the search space and influences adaptation performance. To address this limitation, as illustrated on the right side of Fig. 1, we propose the unrestricted selection of different channel combinations for a chosen rank r_i. Specifically, for the i_{th} trainable low-rank matrix $\mathbf{A}_i \in \mathbb{R}^{r_n\times d}$ and $\mathbf{B}_i \in \mathbb{R}^{d\times r_n}$ with the maximum rank of R, we advocate the freedom to selectively choose channels with the same indices $\mathbf{C}_{r_i}$ to model the incremental update $\Delta\mathbf{W}_i$. This can be formulated as:

$$\Delta\mathbf{W}_i = \mathbf{B}_i[:, \mathbf{C}_{r_i}]\mathbf{A}_i[\mathbf{C}_{r_i}, :] \tag{5}$$

where $\mathbf{C}_{r_i} \in \mathbb{R}^{r_i}$ is sampled from the available indices $\mathbf{C}_{r_n} = [0, 1, \dots, r_n]$ without replacement. This approach allows us to construct a more diverse search space, consequently enhancing the model capacity during the adapter search for downstream task adaptation.

3.3. UCB-inspired channel sampling for supernet training

During the supernet training, we first employ the SPOS algorithm (Guo et al., 2020) to randomly determine a depth denoted as d_i from the high-level depth search space. Given d_i, we then randomly sample a width r_i for each layer from the middle-level width search space. Finally, for each r_i, we utilize the sample-efficient method to select a channel combination denoted as $\mathbf{C}_{r_i}$ from the available choices $\mathbf{C}_{r_n}$ within our low-level search space.

However, as shown in Fig. 2 (left), prior NAS-based adapter methods (Chavan et al., 2023; Liu, 2022) typically fix the channel selection

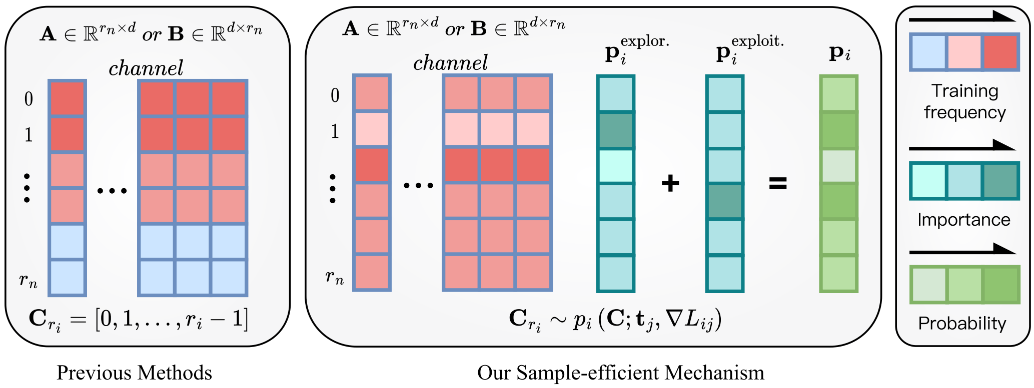


Fig. 2. Our proposed sampling method. Given the selected width r_i, previous methods typically train the top-r_i channels. In contrast, our method calculates the sampling probability $\mathbf{p}_i$ based on the importance distributions of training frequency $\mathbf{p}_i^{\text{exploration}}$ and gradient norms $\mathbf{p}_i^{\text{exploitation}}$. The channels to be trained are then sampled according to $\mathbf{p}_i$.

to the top-r_i indices (i.e., $\mathbf{C}_{r_i} = [0, 1, \ldots, r_i - 1]$) at every training step. This induces a severe training bias: high-index channels are rarely updated, making their true contribution unobservable and compromising the fairness of architecture evaluation.

To address this, we formulate channel selection as a multi-armed bandit (MAB) problem, where each of the C channels (e.g., $C = 768$) corresponds to an arm. Since direct reward measurement is infeasible during supernet training, we use the L_2 norm of per-channel gradients as a proxy for channel utility-larger gradients indicate greater sensitivity to loss reduction and thus higher "reward".

Inspired by the Upper Confidence Bound (UCB) principle (Auer et al., 2002), which balances exploration of uncertain arms and exploitation of high-reward arms, we define a sampling score for each channel $j \in \{1, \ldots, C\}$ as:

$$s_j = g_j + \sqrt{\frac{2 \log T}{t_j + \epsilon}}, \tag{6}$$

where $g_j = \|\nabla_{w_j} \mathcal{L}\|_2$ is the gradient norm of channel j, T is the current global training iteration, t_j is the cumulative number of times channel j has been sampled so far, and $\epsilon = 10^{-6}$ prevents division by zero. The first term encourages exploitation of informative channels, while the second promotes exploration of under-trained ones.

We convert these scores into a valid probability distribution via normalization:

$$p_j = \frac{s_j}{\sum_{k=1}^{C} s_k}, \quad \forall j = 1, \ldots, C, \tag{7}$$

and sample the required r_i channels without replacement according to $\mathbf{p} = [p_1, \ldots, p_C]$.

This mechanism requires no additional hyperparameters beyond standard training settings. In early stages (t_j small for most j), the exploration term dominates, ensuring broad coverage of all channels. As training progresses, t_j increases and the exploitation term prevails, focusing sampling on channels that actively contribute to loss reduction. By dynamically balances exploration and exploitation, we ensure equitable and adaptive training across all candidate channels throughout supernet optimization. Consequently, it enables more reliable evaluation of channel combinations and enhances the generalization performance of the final searched adapters.

In summary, our sampling mechanism is a principled heuristic inspired by UCB. While $p^{\text{exploration}}$ follows bandit theory to balance sampling frequencies, $p^{\text{exploitation}}$ uses gradient norms as a practical proxy for channel contribution to loss reduction.

3.4. Evolutionary search for efficient adapters

Based on the weight-sharing strategy, we inherit the pre-trained supernet weights for candidate adapters to conduct efficient evaluation. We adopt an evolutionary algorithm to search for the optimal adapter. In the first search iteration, we randomly generate candidate adapters that meet the parameter constraints to form the initial population. These models are then evaluated to obtain accuracy by directly adopting the pre-trained supernet weights, and subsequently ranked by accuracies accordingly. The top-ranked adapters are selected for crossover and mutation to produce the next generation. This iterative process allows our method to explore diverse and efficient adapter structures, situated on an improved Pareto curve that balances accuracy and parameter count. In conclusion, the overall procedure of our proposed SEA is summarized in our supplementary material.

3.5. Inference analysis

Due to the absence of non-linear functions in the hierarchical adapters for each pre-trained Linear, these adapters can be seamlessly incorporated into the pre-trained weight $\mathbf{W}^{(0)}$ after the search. Suppose we ultimately discover an adapter with m layers, substituting all intermediate outputs $\boldsymbol{y}_0$ to $\boldsymbol{y}_{m-1}$ into the final output $\boldsymbol{y}_m$ and we can obtain the following output:

$$\begin{aligned} \boldsymbol{y}_m &= \boldsymbol{x}\mathbf{W}^{(0)}(\mathbf{1} + \mathbf{W}_m)[(\prod_{k=0}^{m-1} \mathbf{W}_k + \prod_{k=1}^{m-1} \mathbf{W}_k + \ldots + \mathbf{1}) \\ &\quad + ((\prod_{k=1}^{m-1} \mathbf{W}_k)\mathbf{b}_0 + (\prod_{k=2}^{m-1} \mathbf{W}_k)\mathbf{b}_1 + \ldots + \mathbf{b}_{m-1})] \\ &\quad + \mathbf{b}^{(0)}(\mathbf{b}_m + \mathbf{1}) = \boldsymbol{x}\mathbf{W}^{\star} + \mathbf{b}^{\star} \end{aligned} \tag{8}$$

Therefore, we can pre-compute the inference weights $\mathbf{W}^{\star}$ and $b^{\star}$ for downstream tasks using the discovered adapter and the pre-trained model weights. These inference weights can then be directly loaded into the original pre-trained model, enabling efficient inference without incurring extra computational overhead.

4. Experiments

In this section, we present a systematic evaluation of SEA across three adaptation scenarios, ordered by increasing task complexity: natural language understanding (GLUE), image classification (VTAB and ImageNet), and dense prediction (COCO detection and segmentation). We first describe the experimental setup, followed by results and analysis for each task category. Additional experiments and ablation analyses are provided in the supplementary material.

4.1. Experimental settings

Pre-trained backbone. We employ the pre-trained Vision Transformer (Dosovitskiy et al., 2021) to adapt to downstream tasks.

Table 1
Text classification results on MNLI and SST2. $\mathbf{P}_{train}$ and $\mathbf{P}_{test}$ denote the parameters introduced by fine-tuning.

Method	$\mathbf{P}_{train}$ (M)	$\mathbf{P}_{test}$ (M)	MNLI (%)	SST2 (%)
Full (Chavan et al., 2023)	125	**0**	88.0	95.0
MAM-Adapter (He et al., 2022a)	46.8	0.6	87.4	94.2
Res-Tuning (Jiang et al., 2023)	**1.0**	1.0	87.5	94.6
SEA	1.6	**0**	**88.8**	**96.1**

Experiments are conducted using the pre-trained ViT-B/16 on ImageNet-21k (Deng et al., 2009) for downstream datasets including CIFAR-100 (Krizhevsky et al., 2009), VTAB-1k (Zhai et al., 2019), ImageNet, and others. For dense prediction tasks on the COCO (Lin et al., 2014) dataset, we adopt the Mask R-CNN (He et al., 2017) with the pre-trained backbone of Swin-Transformer-Tiny (Liu et al., 2021).

Baselines. Our experimental comparisons involve the following three categories of methods: (i) According to GLoRA (Chavan et al., 2023), we set two traditional fine-tuning methods, including full training and linear probing. The former fine-tunes the entire set of pre-trained model parameters, while the latter only fine-tune the network output head; (ii) Hand-crafted Adapters, such as the introduction of learnable token embedding method VPT, the inclusion of serial branch fine-tuning methods Adapter (Houlsby et al., 2018) and SSF (Lian et al., 2022), and the introduction of parallel fine-tuning branch methods Side-Tuning (Zhang et al., 2020), LST (Sung et al., 2022), LoRA (Hu et al., 2022), AdaptFormer (Chen et al., 2022), and Res-Tuning (Jiang et al., 2023); (iii) NAS-based Adapters, including NOAH (Liu, 2022) and GLoRA (Chavan et al., 2023).

Implementation Details. SEA consists of two stages for downstream task adaption: supernet training and adapter search. At the first stage, we designate a searchable adapter for each Linear layer within the Transformer Block, setting the adapter depth to $D = \{1, 2, 3\}$, with searchable rank sizes from $R = \{2, 4, 8\}$ for each depth. For each selected rank r, channel combinations C can be freely chosen from the maximum rank, resulting in an extensive search space. For instance, for a ViT-B/16 with 12 layers and 4 Linear layers in each block, there are approximately $[\sum_{k=1}^{3}(C_8^2 + C_8^4 + C_8^8)^k]^{48} \approx 10^{288}$ possible adapters in our search space. The detailed configurations of the supernet training are provided in subsequent sections. Regarding the adapter search, we set the maximum number of generations to 20. In the initial phase, 50 candidate adapters are randomly generated. Then each subsequent generation produces 25 models through mutation and crossover based on the top 10 ranked models from the previous generation. The best-performing adapter is selected as our final result.

4.2. Adaption on language understanding

We conduct experiments on the SST2 (Socher et al., 2013) and MNLI (Williams et al., 2018) datasets for text classification. SST2 is a sentiment analysis dataset for movie reviews, while MNLI is designed to understand the logical relationships between texts. We adopt the pre-trained RoBERTa (Liu et al., 2019b), which is based on the $\text{BERT}_{\text{BASE}}$ (Kenton & Toutanova, 2019) architecture. SEA searches for adapters within each linear layers of the Transformer. (Table 1)

Res-Tuning achieves only 0.1% and 0.4% improvement over MAM-Adapter on MNLI and SST2, respectively, and falls short of full training. While SEA introduces an additional 0.6M parameters compared to Res-Tuning, it gets 1.3% and 1.5% accuracy improvements on the two datasets, **even surpassing the performance of full fine-tuning**. Due to the design principles of SEA, it incurs no additional inference parameters and thus more efficient.

4.3. Adaption on image classification

Transfer Learning. We evaluate our approach on the VTAB-1k benchmark by conducting transfer experiments on 19 diverse visual datasets using the pre-trained ViT-B/16. VTAB-1k comprises datasets categorized into three main types: natural, specialized, and structured, sourced from standard cameras, non-standard cameras, and synthetic images, respectively. Each dataset consists of 1000 training samples, making it exceptionally challenging.

We present a comprehensive comparison with other adapters in Table 2. SEA obtains the highest "Group Mean" accuracy 78.50 across all 19 datasets, achieving the best in 11 datasets, and the second-highest accuracy in 4 of them. The intuitive comparison of line charts on different tasks is shown in Fig. 3.

Compared to the latest manually designed adapter Res-Tuning (Jiang et al., 2023), SEA achieves an average accuracy improvement of over 2%. SEA also outperforms NAS-based adapters NOAH (Liu, 2022) and GLoRA (Chavan et al., 2023) by 3% and 0.5%, respectively. For the structured data, SEA does not perform as well as Natural and Specialized. However, SEA still achieves the highest mean accuracy on this group of datasets. **Despite using slightly more parameters, SEA matches the throughput of the original pre-trained model and surpasses most other methods. These improvements might stem from our hierarchical search space design and the sample-efficient mechanism**, which promotes an effective exploration in the extensive search space to accommodate diverse downstream datasets, thus achieving superior performance in various scenarios.

Few-Shot Learning. We evaluate SEA with limited training data through few-shot learning. Following previous studies, we use five visual recognition datasets: Food101 (Bossard et al., 2014), OxfordFlowers102 (Nilsback & Zisserman, 2008), OxfordPets (Parkhi et al., 2012), StandfordCars (Gebru et al., 2017), and FGVCAircraft (Maji et al., 2013). In comparison to other state of the arts, we train the supernet on 1, 2, 4, 8, and 16 shots and then search for the best adapter. We evaluate related methods on identical dataset splits to ensure a fair comparison. Results are illustrated in Fig. 4. SEA fulfills a remarkable superiority over existing approaches across almost all datasets, with 5 different shots. As the number of training samples increases, the effectiveness of fine-tuning methods has shown gradual improvement, and SEA continues to maintain a leading edge. This achievement affirms the effectiveness of our method.

Domain Generalization. We conduct domain generalization experiments on the ImageNet (Deng et al., 2009) dataset to test the robustness of SEA by evaluating the model's ability to recognize Out-Of-Distribution (OOD) samples. The supernet is trained on 16 shots for each class in the source domain ImageNet-1k and then performs the adapter search on the validation set. Without additional fine-tuning, we leverage the discovered adapter for zero-shot evaluation on the target domains ImageNet-A (Hendrycks et al., 2021b), ImageNet-R (Hendrycks et al., 2021a), ImageNet-V2 (Recht et al., 2019), and ImageNet-Sketch (Wang et al., 2019). The experimental results are presented in Fig. 5.

Results in Fig. 5 demonstrate the robustness of our method in domain shift scenarios. SEA attains the highest accuracy in the source domain and the first three target domains. Despite SEA having lower accuracy on ImageNet-Sketch than GLoRA, it still obtains the second-highest accuracy, also showing that the performance across target domains is not absolutely positively correlated. Finally, SEA fulfills the highest average accuracy of 36.0%, showcasing its robustness and out-of-distribution capabilities.

4.4. Adaption on dense prediction

By leveraging the SEA method, we adapt the pre-trained Swin-Transformer-Tiny (Liu et al., 2021) on the COCO dataset

Table 2
Transfer learning results on the VTAB benchmark. Values in the table represent accuracy percentages. "Group mean" refers to the average of the mean results for the three sub-groups. "Throughput" indicates the number of processed images per second on a NVIDIA 3080 GPU with a batch size of 128. **Bold** font represents the best performance, while underlined font indicates the second highest accuracy (Kim et al., 2024).

Dataset	Natural							Specialized				Structured										
	CIFAR-100	Caltech 101	DTD	Flowers 102	Pets	SVHN	Sun 397	Camelyon	EuroSAT	Resisc 45	Retinopathy	Clevr-Count	Clevr-Dist	DMLab	KITTI-Dist	dSpr-Loc	dSpr-Ori	sNORB-Azim	sNORB-Elev	Group Mean	Throughout	Param. (M)
Traditional Methods																						
Full (Chavan et al., 2023)	68.9	87.7	64.3	97.2	86.9	87.4	38.8	79.7	95.7	84.2	73.9	56.3	58.6	41.7	65.5	57.5	46.7	25.7	29.1	68.97	589	85.84
Linear (Chavan et al., 2023)	63.4	85.0	63.2	97.0	86.3	36.6	51.0	78.5	87.5	68.6	74.0	34.3	30.6	33.2	55.4	12.5	20.0	9.6	19.2	57.64	589	0.04
Hand-crafted Adapters																						
Side-Tuning (Zhang et al., 2020)	60.7	60.8	53.6	95.5	66.7	34.9	35.3	58.5	87.7	65.2	61.0	27.6	22.6	31.3	51.7	8.2	14.4	9.8	21.8	49.91	<589	9.59
Adapter (Houlsby et al., 2018)	74.2	85.7	62.7	97.8	87.2	36.4	50.7	76.9	89.2	73.5	71.6	45.2	41.8	31.1	56.4	30.4	24.6	13.2	22.0	60.52	<589	1.82
LST (Sung et al., 2022)	58.0	87.1	66.2	99.1	89.7	63.2	52.6	81.9	92.2	78.5	69.4	68.6	56.1	38.8	73.4	72.9	30.5	16.6	31.0	67.56	<589	0.89
VPT-Deep (Jia et al., 2022)	**78.8**	90.8	65.8	98.0	88.3	78.1	49.6	81.8	96.1	83.4	68.4	68.5	60.0	46.5	72.8	73.6	47.9	32.9	37.8	71.97	422	0.60
LoRA (Hu et al., 2022)	67.1	91.4	69.4	98.8	90.4	85.3	54.0	84.9	95.3	84.4	73.6	82.9	**69.2**	49.8	78.5	75.7	47.1	31.0	44.0	74.60	589	0.29
AdaptFormer (Chen et al., 2022)	70.8	91.2	70.5	99.1	90.9	86.6	54.8	83.0	95.8	84.4	76.3	81.9	64.3	49.3	80.3	76.3	45.7	31.7	41.1	74.75	555	0.16
SSF (Lian et al., 2022)	69.0	92.6	75.1	99.4	91.8	90.2	52.9	87.4	95.9	87.4	75.5	75.9	62.3	53.3	80.6	77.3	54.9	29.5	37.9	75.69	589	0.24
Res-Tuning (Jiang et al., 2023)	75.2	92.7	71.9	99.3	91.9	86.7	**58.5**	86.7	95.6	85.0	74.6	80.2	63.6	50.6	80.2	**85.4**	**55.7**	31.9	42.0	76.32	<589	0.55
NAS-designed Adapters																						
NOAH (Liu, 2022)	69.6	92.7	70.2	99.1	90.4	86.1	53.7	84.4	95.4	83.9	75.8	82.8	68.9	49.9	81.7	81.8	48.3	32.8	**44.2**	75.48	535	0.42
Hydra (Kim et al., 2024)	72.7	91.3	72.0	99.2	91.4	90.7	55.5	85.8	96.0	86.1	75.9	**83.2**	68.2	50.9	82.3	80.3	50.8	34.5	43.1	76.50	589	0.28
GLoRA (Chavan et al., 2023)	76.4	92.9	74.6	**99.6**	92.5	91.5	57.8	87.3	96.8	88.0	76.0	83.1	67.3	54.5	**86.2**	83.8	52.9	37.0	41.4	77.97	589	0.86
SEA	77.6	**93.2**	**75.2**	**99.6**	**93.0**	**92.4**	57.8	**88.8**	**96.9**	**88.3**	**76.8**	82.2	67.6	**55.2**	85.4	83.3	54.9	**37.7**	43.2	**78.50**	589	1.49

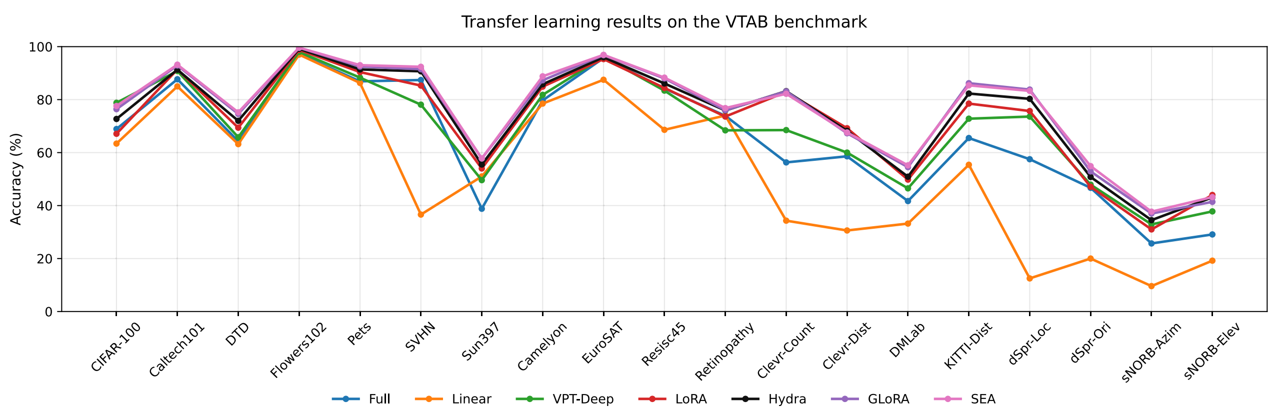


Fig. 3. VTAB transfer learning performance across all 19 tasks. Methods are evaluated on natural, specialized, and structured task groups. Our method **SEA** consistently achieves competitive or state-of-the-art results.

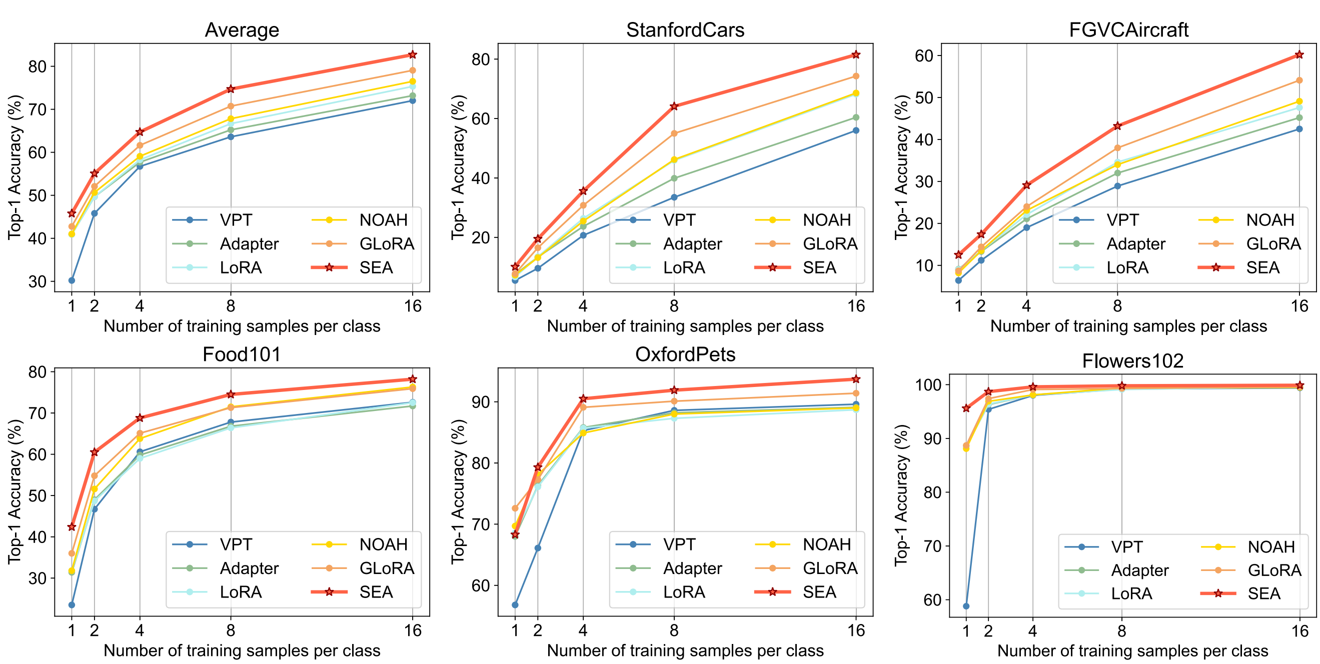


Fig. 4. Results for few-shot learning on five datasets. Average accuracy is provided in the top-left corner.

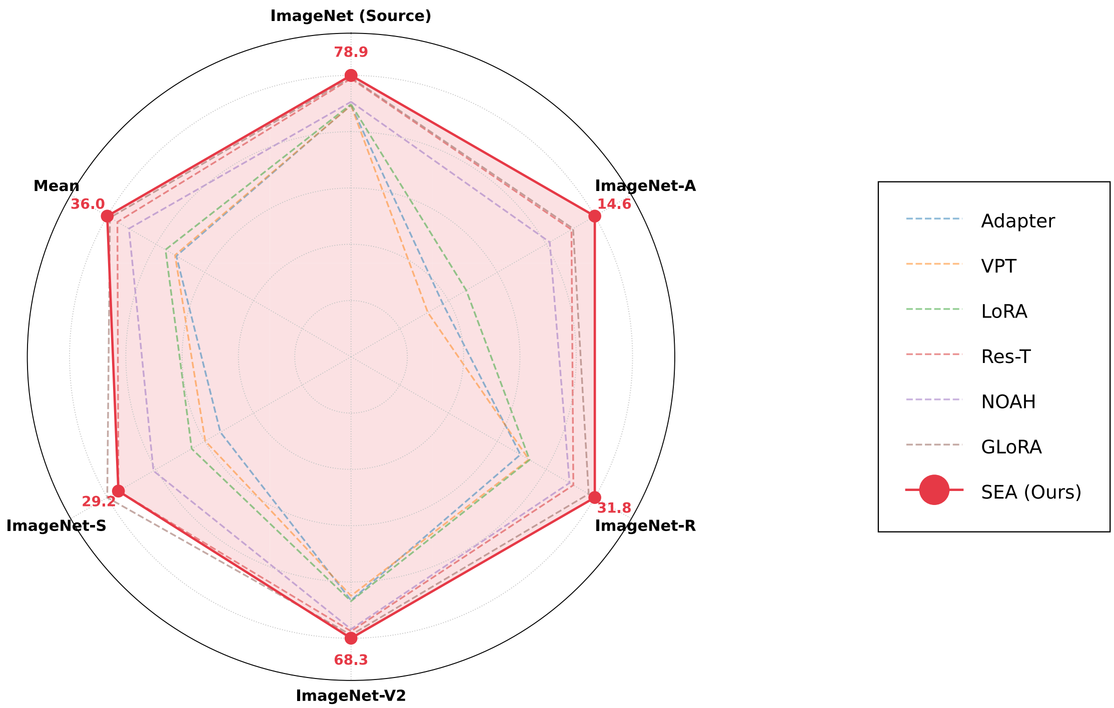


Fig. 5. Domain generalization results on the ImageNet dataset.

Table 3
Detection and segmentation results on COCO. † and ⋆ distinguish between Swin-Transformer-Tiny pre-trained on the more general ImageNet-21k dataset and the more object-centric object-365 dataset, while others are pre-trained on ImageNet-1k.

Method	Param.	AP^b	AP^b_{50}	AP^b_{75}	AP^m	AP^m_{50}	AP^m_{75}
Full (Lian et al., 2022)	47.8	**43.7**	**66.6**	**47.7**	**39.8**	**63.3**	**42.7**
Linear (Lian et al., 2022)	20.3	31.7	55.7	32.5	31.2	53.0	32.2
VPT (Jia et al., 2022)	-	33.8	57.6	35.3	32.5	54.5	33.9
SSF (Lian et al., 2022)	21.6	34.9	58.9	36.1	33.5	55.8	34.7
SEA	**21.5**	36.3	59.7	38.9	35.0	56.6	36.2
	24.1	38.4	61.8	41.0	36.0	58.7	38.1
SEA†	24.1	40.0	63.6	42.9	37.2	60.3	39.5
SEA⋆	24.1	42.2	65.3	46.5	39.1	62.1	41.9

(Lin et al., 2014) for object detection and instance segmentation. COCO contains a variety of common objects in context, including 118k training images and 5k validation images. We evaluate the performance using average precision for both tasks. (Table 3)

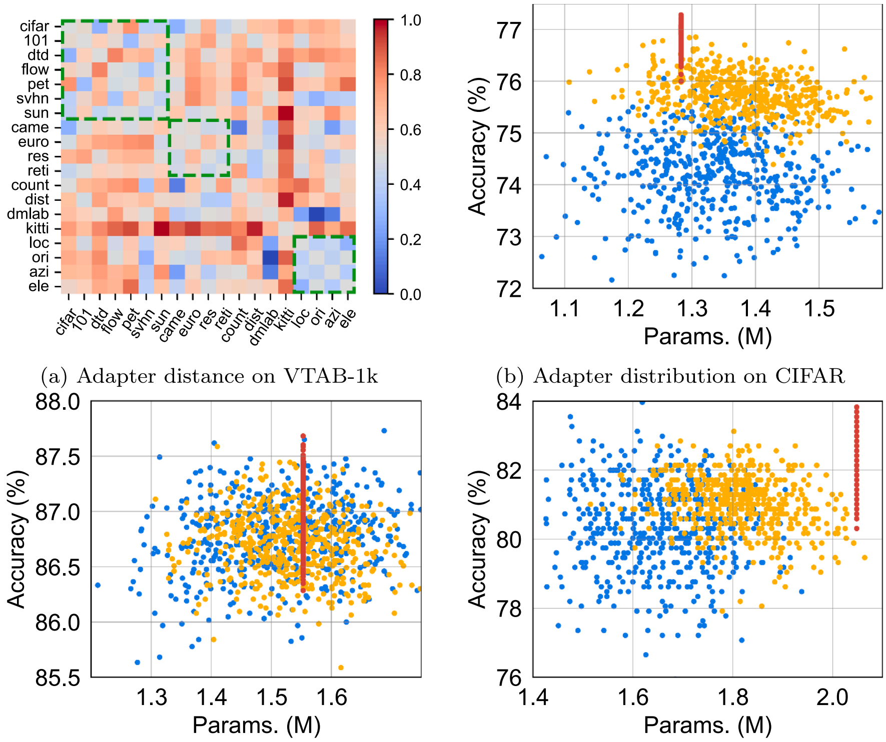
Full training includes 20.3M parameters for task-specific heads and 27.5M backbone parameters, leading to 47.8M parameters in total. With only approximately 1M backbone parameters fine-tuned, SEA achieves higher detection and segmentation accuracy than SSF (Lian et al., 2022), but such results still lags behind full training. To bridge this gap, we explore three strategies: increasing trainable backbone parameters to 3.8M, using stronger backbones pre-trained on ImageNet-21K (SEA†) and a task-relevant backbone pre-trained on Object365 (Shao et al., 2019) (SEA⋆). These approaches progressively narrow the performance gap, with SEA⋆ achieving performance on par with full training. **This suggests that selecting pre-trained models closely aligned with downstream tasks can significantly enhance transfer performance**.

4.5. More in-depth analysis

Distance of searched adapters. We convert each discovered adapter structure into binary encoding and then measure the Hamming distance to compute the differences between them. Fig. 6(a) shows the results on 19 datasets of VTAB-1k. We can observe that most squares appear in deep red, indicating significant differences in the optimal adapter structures found across different datasets, **highlighting the necessity of using NAS for adapter design**. However, three light blue areas within the green dashed boxes emerge along the diagonal, indicating similarities in these adapters. They exactly correspond to the three types of datasets in VTAB-1k (Natural, Specialized, and Structured), suggesting that similar downstream datasets may potentially share optimal adapter structures. These observations may provide valuable insights for future adapter design.

Adapter distribution in the search space. To analyze the effectiveness of different granularities in the search space, we randomly search for 500 adapters near the optimal adapter, allowing only one dimension to be searched at a time. Figs. 6(b), (c), and (d) present the results on three datasets. Depth search tends to yield more widely distributed adapters, while width search further approaches the local optima based on the depth search. Furthermore, channel combination search significantly improves the adapter performance, building upon the two aforementioned search granularities. These distributions **illustrates a hierarchical adapter evolution, demonstrating the effectiveness of our proposed hierarchical search space**. SEA not only supports searching for diverse adapters but also enables adapters to continuously evolve for downstream tasks.

Applicability and Limitations. Low-rank adapters LoRA (Hu et al., 2022) claimed in the introduction, are designed under the premise that **a powerful pretrained model already provides the necessary representations for downstream tasks, and only minor task-specific adjustments are required**. Consequently, their effectiveness hinges on the alignment between the pretrained features and the target task. For instance, ImageNet-21k pretrained ViT-B/16 offers sufficiently rich and generalizable features for image classification benchmarks like VTAB-1k and CIFAR-100, enabling adapters to achieve strong performance despite their constrained update subspace. In contrast, when using an ImageNet pretrained Swin Transformer optimized solely for image-level classification-for COCO dense prediction, the pretrained features lack essential spatial and instance-level priors. **Adapters have to reconstruct detection/segmentation capabilities from classification-**



(a) Adapter distance on VTAB-1k (b) Adapter distribution on CIFAR

(c) Adapter distribution on Resisc (d) Adapter distribution on KITTI

Fig. 6. Adapter distance and adapter distributions. (a): We measure distances of 19 discovered adapters on VTAB-1k and show the normalized distances, where deeper colors indicate larger distances. Distances on the diagonal is set to gray to avoid influencing the observation. (bcd): Distributions of adapters in our hierarchical search space in terms of parameters and accuracy. Blue dots represent depth-only search, yellow dots represent width-only search, and red dots represent channel combination-only search.

oriented features, which exceeds the capacity of low-rank updates. However, with Objects365 pretraining-which inherently embeds object-centric representations closely aligned with COCO-the gap between pre-trained features and downstream requirements narrows significantly, allowing adapters to operate efficiently.

SEA adheres to the same scope and limitations of adapters' methods. Further, SEA aims to identify the optimal adapter configuration under a given low-rank budget-not to unconditionally surpass full fine-tuning. By constructing a hierarchical search space over width, depth, and rank, and by adaptively composing structures via evolution-guided search, SEA expands the expressivity of conventional fixed-form adapters. As evidenced by our experiments, SEA consistently outperforms existing adapters within their shared regime of applicability: namely, when the pretrained model provides adequate task-relevant features. In such settings, SEA enables more flexible and efficient adaptation; otherwise, full fine-tuning remains preferable.

5. Conclusion

In this work, we propose SEA (Searchable Efficient Adapter), a neural architecture search framework for parameter-efficient fine-tuning that advances existing NAS-based approaches through a systematically decoupled search space. Unlike prior methods that jointly optimize adapter type and width or rely on fixed architectural templates, SEA explicitly disentangles key design dimensions, enabling more flexible and fine-grained adaptation strategies. This structured decomposition, combined with a UCB-guided training mechanism to mitigate evaluation bias during supernet optimization, allows SEA to discover task-specialized configurations that generalize robustly across diverse domains. As demonstrated on GLUE, VTAB, ImageNet, and COCO, our method consistently outperforms both hand-crafted adapters and recent NAS-designed counterparts.

Nevertheless, our approach has limitations. First, while SEA reduces the need for task-specific parameters at test time, the evolutionary search phase remains computationally demanding, limiting its direct applicability to extremely large-scale models (e.g., Llama-3 or ViT-G) without further acceleration. Second, although SEA demonstrates strong transferability across backbones (ViT, Swin, RoBERTa), its search space is currently tailored to linear-style adapters; extending it to non-linear or dynamic architectures remains an open challenge.

Looking forward, we envision two promising directions: (1) *Efficient search*: integrating gradient-based NAS or one-shot distillation to replace evolutionary search; (2) *Cross-modal adaptation*: applying SEA to unified foundation models (e.g., Flamingo, LLaVA) for joint vision-language tuning;

We hope SEA not only advances practical efficiency in model adaptation but also inspires a broader rethinking of *how we train*, not just *what we design*, in the era of foundation models.

CRediT authorship contribution statement

Shun Lu: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization; **Fangyuan Mao:** Visualization, Methodology, Data curation; **Junkun Chen:** Writing – review & editing, Validation, Software; **Jilin Mei:** Writing – review & editing, Supervision, Investigation, Funding acquisition; **Chen Min:** Writing – review & editing; **Yan Xing:** Writing – review & editing, Resources, Funding acquisition; **Yu Hu:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Yu HU reports financial support was provided by National Natural Science Foundation of China (NSFC). Jilin Mei reports financial support was provided by National Natural Science Foundation of China (NSFC). Yu Hu reports a relationship with National Natural Science Foundation of China (NSFC) that includes: funding grants. Jilin Mei reports a relationship with National Natural Science Foundation of China (NSFC) that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Overview

In the subsequent sections, we delve into a comprehensive analysis and discussion of the design of SEA, as outlined in Appendix B. This is intended to enhance the understanding of our method. Further, the detailed search process is provided in Appendix C. Finally, to underscore the versatility of our approach, Appendix D presents additional experimental comparisons, illustrating its robust generalization capabilities across a variety of downstream tasks.

Table A.4
Number of block variants and size of the search space.

Method	Block Variants	Search Space
NOAH (Liu, 2022)	80	$80^{12} \approx 10^{23}$
GLoRA (Chavan et al., 2023)	483^4	$483^{48} \approx 10^{129}$
SEA	980199^4	$980199^{48} \approx 10^{288}$

Appendix B. Analysis and discussion

Analysis on the size of the search space. Table A.4 summarizes the size of different search spaces using the pre-trained ViT-B/16 with 12 Transformer blocks. NOAH (Liu, 2022) searches for the position and length of VPT (Jia et al., 2022), as well as the position and width of Adapter (Houlsby et al., 2018) and LoRA (Hu et al., 2022). Each layer has a maximum of 80 distinct blocks. Moreover, the position search is constrained to layers in multiples of three (i.e. 3, 6, 9, or 12 layers), resulting in a relatively compact search space with less than 80^{12} possible variants. GLoRA searches for unique learnable matrices for every Linear transformation, offering 483 distinct configurations for each pre-trained Linear. Given that each block contains four Linear layers (a QKV transformation, a Projection layer, and a two-layer FFN), this results in 483^4 variants per block, expanding the search space to an immense 483^{48} adapters, significantly larger than that of NOAH.

SEA further diversifies the adapter structures by jointly searching for the depth, width, and channel combination of each Linear transformation. With $[\sum_{k=1}^{3}(C_8^2 + C_8^4 + C_8^8)^k] = 980,199$ variants per Linear and $980,199^4$ possible structures per block, SEA constructs an even more extensive search space of $980,199^{48}$ with 12 searchable layers, enabling adaptive and task-specific searches for the most suitable adapters for downstream tasks. This enhancement significantly boosts the transferability of adapters across various tasks, as shown in previous experimental comparisons and Appendix D.

Analysis on the search cost and inference efficiency. Although the number of parameters of our searched adapters is slightly larger than previous works, our method *introduces only a marginal increase in training overhead while keeping efficient during inference*. SEA adopts a two-stage search for adapters: supernet training and evolutionary search. Taking the VTAB-1k (Zhai et al., 2019) benchmark as an example, the average supernet training time across 19 datasets is 2.2 hours on one NVIDIA Tesla V100 GPU, nearly identical to that of GLoRA (Chavan et al., 2023) and comparable to hand-crafted adapters. The primary increase in time comes from the search phase, which can be largely accelerated through parallelization. For instance, when employing an 8-card parallel search, SEA requires an average of just 2.3 hours to complete the search process. In terms of inference, our searched adapters can be fully integrated into the pre-trained model, and thus do not incur additional overhead due to the design principles of our search space, regardless of the number of parameters. This advantage makes our method even more efficient compared to some manually designed adapters such as VPT (Jia et al., 2022), and AdaptFormer (Chen et al., 2022).

Table B.5
Ablation study for our sampling method.

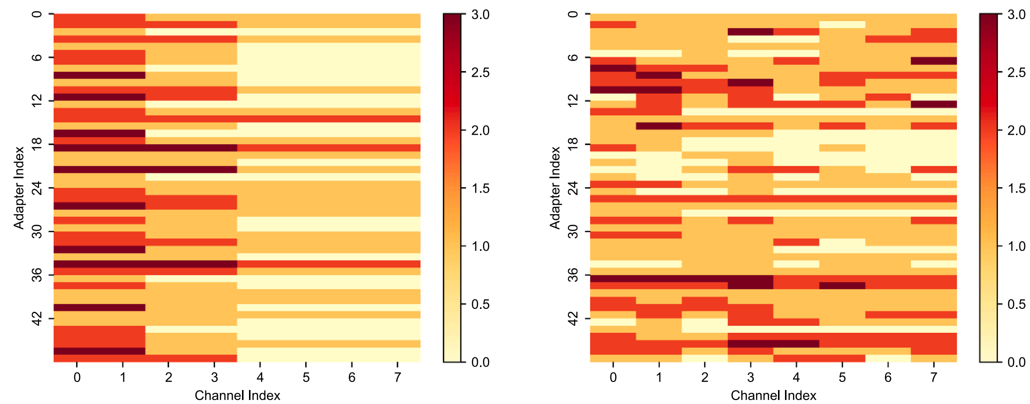
Method	Exploitation	Exploration	Acc. (%)
SPOS (Guo et al., 2020)	-	-	93.58
SEA	-	✓	93.72
	✓	-	93.85
	✓	✓	**94.09**

The impact of the biased training. To reveal the impact of the biased training, we compare the frequency of the selected channels in searched adapters using traditional methods with those via SEA within the same depth and width search space on the CIFAR-100 dataset. For each adapter, we compute the average selected frequency of each channel and depict the results in Fig. B.7.

Favoring top-channels (Chavan et al., 2023; Liu, 2022) during the supernet training leads to a biased training outcome, resulting in a predilection towards selecting top-channels for the searched adapters. This biased selection, as indicated in Fig. B.7(a), typically culminates in a classification accuracy of only 93.53%. In contrast, Fig. B.7(b) reveals that SEA displays varying preferences for different channels across different layers, with a more balanced channel selection in the intermediate layers, while the other layers exhibit distinct preferences. SEA achieves a much higher accuracy of 94.09%, substantiating the efficacy of our designed channel combination search and the UCB sampling approach.

Ablation study for the sampling method. To investigate the importance of different components in the supernet training, Table B.5 shows the ablation results on the CIFAR-100 dataset. When both exploration and exploitation are disabled, the adapter's accuracy degrades to 93.58%. However, when either of the two components is enabled, the performance of the searched adapter improves. Moreover, excessive emphasis on exploration does not benefit the adapter search. When both components are combined, early emphasis on exploration and later focus on exploitation produce the best adapter with an accuracy of 94.09%.

Robustness of UCB-inspired Sampling. While gradient norms are inherently mini-batch-dependent, our UCB-inspired sampling mechanism remains stable in practice due to the exploration-exploitation balance. In early training stages, all adapter candidates have low selection counts, so the exploration term (Eq. 6) dominates-effectively yielding near-uniform sampling akin to SPOS-and ensures every candidate receives sufficient initial updates. Concurrently, we use **gradient clipping to prevent abrupt norm fluctuations**, rendering the exploitation term (Eq. 7) negligible initially. As training progresses, **gradient norms are accumulated over batches (e.g., via running averages)**, smoothing out mini-batch noise and enabling reliable ranking only after sufficient statistics are gathered. This delayed, averaged feedback prevents premature convergence or degenerate behavior (e.g., permanently ignoring viable adapters). Empirically, we observe stable sampling dynamics across both few-shot and dense prediction tasks. Should higher sensitivity arise in other settings, simple enhancements-such as EMA smoothing



(a) Frequency of selected channels in adapters by previous methods

(b) Frequency of selected channels in adapters by SEA

Fig. B.7. The frequency of the selected channels in searched adapters. We analyze the average channel frequency for each adapter. Figure (a) shows the channel frequency in the searched adapters of previous methods. Top-channels are predominantly chosen across nearly all adapters. Figure (b) displays the results within adapters by SEA, indicating a distinct layer-specific preference for varying channels.

of gradient norms or scheduled rebalancing of exploration/exploitation weights (e.g., linear decay)-can further enhance robustness.

Appendix C. Search algorithm

To facilitate better comprehension of our adapter search process, we summarize the detailed steps in Algorithm 1.

Algorithm 1 SEA.

Input: Downstream training data $\mathbb{D}_T$, validation data $\mathbb{D}_V$, supernet $\mathcal{S}$ with pre-trained weights W and adapter weights A and B, total training iterations T_0, training steps per epoch T_e, total search populations T_1.
Output: The best adapter with fine-tuned parameters.

```
for i = 1 to T_0 do                                          ▷ Supernet Training
    Randomly sample the depth and rank for adapters;
    Sample the channels based on the distribution p;
    Randomly sample a mini-batch data from D_T;
    Train adapter weights A and B by gradient descent;
    Update p^exploration according to Eq. 6;
    Record gradient norms of the sampled channels;
    if i % T_e = 0 then
        Update p^exploitation according to Eq. 7;
    end if
    Update sampling distribution p according to Eq. 8
end for
for j = 1 to T_1 do                                          ▷ Adapter Search
    if j = 1 then
        Randomly generate candidate adapters;
    end if
    Evaluate and rank candidate adapters on D_V;
    Apply mutation to top-performing adapters;
    Apply crossover to top-performing adapters;
end for
Incorporate searched adapters to the pre-trained model.
```

Appendix D. Additional adaption experiments

Adaption on Image Classification. To further validate the transferability of our method in downstream tasks, we follow AdaptFormer (Chen et al., 2022) to employ the pre-trained ViT-B/16 to adapt to two types of datasets, SVHN (Goodfellow et al., 2013) and Food-101 (Bossard et al., 2014). The Street View House Numbers (SVHN) dataset is a real-world digit recognition dataset, comprising 10 classes with 73,257 training samples and 26,032 test samples. Food-101, on the other hand, encompasses 101 different food categories, each containing 750 training and 250 test samples.

The experimental results, as shown in Table D.6, indicate that SEA, with a slightly larger parameter size of 0.12M than AdaptFormer, achieves superior accuracy on both datasets, outperforming AdaptFormer by 0.24% and 0.19% respectively. Specifically, on SVHN, SEA attains an accuracy of 97.53% using only 1.6% of the parameters required for Full tuning. This performance surpasses other fine-tuning methods like VPT and AdaptFormer. On Food-101, SEA achieves the highest accuracy of 91.08%, becoming the first method to exceed the Full tuning paradigm.

Table D.6
Adaption results with ViT-B/16 on SVHN and food-101. The best accuracy among all methods except full tuning is **bolded.** †: We report the average parameters of the searched adapters on two datasets for SEA.

Method	Params. (M)	SVHN	Food-101
Full	85.90	95.41	90.96
Linear	**0.07**	55.36	88.14
VPT (Zheng et al., 2021)	0.08	92.77	90.16
AdaptFormer (Chen et al., 2022)	1.26	97.29	90.89
SEA	1.38†	**97.53**	**91.08**

Adaption on Image Segmentation. We conduct the semantic segmentation adaption experiment on the ADE20K dataset, which encompasses over 20k scene-centric images. It spans a diverse range of 150 categories, including stuff like skies, roads, and grasses, as well as discrete objects such as persons, cars, and beds. This dataset contains 20k training samples and 2k validation samples. We employ the SETR (Zheng et al., 2021) segmentation framework based on the pre-trained ViT-L/14 and report the mean Intersection over Union (mIoU) on the validation set.

We depict the comparison of different fine-tuning paradigms in Table D.7. Due to the inherent misalignment between the objectives of pre-training and downstream tasks, this adaptation becomes a hard nut to crack. For example, fine-tuning the segmentation head results in an inferior mIoU of 35.12 that pales significantly in comparison to the full-tuning result of 48.31. Moreover, recent work RepAdapter (Luo et al., 2023) also obtained marginal improvement on this task, achieving a 0.4% increase over the previous adapter approach VPT. However, SEA

Table D.7
Comparison of ADE20k validation results with SETR (Zheng et al., 2021) using the pre-trained ViT-L (Zhou et al., 2019). The best mIoU score among all methods except full tuning is **bolded**.

Method	Params. (M)	mIoU
Full tuning (Jia et al., 2022)	318.31	48.31
Head only (Jia et al., 2022)	**13.18**	35.12
Bias (Zheng et al., 2021)	13.46	43.40
VPT (Zheng et al., 2021)	13.43	42.11
VPT + Bias (Zheng et al., 2021)	15.79	44.04
RepAdapter (Luo et al., 2023)	13.82	44.44
SEA	22.89	**45.48**

surpasses the previously top-tier RepAdapter by 1.04%, achieving a remarkable enhancement mIoU of 45.48%, setting a new record in this adaption task and also taking a significant step towards approaching the results of Full tuning. Despite SEA employing larger parameters compared to previous efforts, these parameters are only 7.2% of those in full tuning. More importantly, they can be fully integrated into the pretrained model, ensuring no additional overhead during inference.

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